

Fine-Tuning Over Architectural Complexity: Broad-Coverage PII Detection on PIIBench with DeBERTa

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<https://github.com/pritesh-2711/pii-bench>

Abstract. Personally identifiable information (PII) detection systems are frequently trained within narrow source or domain boundaries, limiting coverage when deployed on heterogeneous text. We study model fine-tuning on a corrected multi-source PIIBench preparation spanning 82 retained entity types across ten source datasets. We evaluate three DeBERTa-based approaches: direct token-classification fine-tuning, a source-conditioned hierarchical model (SC+H), and a three-phase curriculum extension (SC+H+Curr). Against eight published comparator systems on a reproducible 5,000-record held-out subset (test_5k), direct fine-tuned DeBERTa achieves F1 0.6476, while SC+H and the curriculum variant achieve 0.5899 and 0.2772 respectively; the strongest published comparator reaches only 0.1723. Because validation initially favoured SC+H, we perform a final streamed evaluation on the complete 100,002-record held-out split. Direct fine-tuning remains superior, achieving F1 0.6455 versus 0.5894 for SC+H. Entity-level analysis shows that direct fine-tuning wins 54 of 82 fine entity types and all ten coarse groups by support-weighted entity F1, while SC+H retains localised advantages on 28 types. The results indicate that diverse task-specific training data and a simple weighted cross-entropy objective contribute more to broad-coverage PII detection than the tested architectural and curriculum complexity.

Keywords: PII detection, named entity recognition, DeBERTa, fine-tuning, multi-source benchmarks, curriculum learning, token classification

1. Introduction

The detection and redaction of personally identifiable information (PII) is a prerequisite for regulatory compliance under GDPR, HIPAA, and CCPA. Production systems processing user-generated or enterprise text must identify spans corresponding to sensitive entities—names, contact details, financial identifiers, credentials—and protect them before downstream use. Despite the practical importance, no single published model or rule-based engine covers the full taxonomy of PII types that appear across real-world corpora.

PIIBench is a unified benchmark consolidating ten heterogeneous source datasets into a standardised BIO token-classification task. Evaluating eight published systems on a stratified held-out subset, the original paper established that all systems achieve span-level F1 below 0.14, directly quantifying the domain-silo problem: each system covers at most a small slice of the full PII taxonomy it encounters in deployment.

Against this backdrop, the performance gap is stark. **The best published system on our corrected dataset benchmark achieves span-level F1 of 0.1723 (SpanMarker BERT). Our direct fine-tuned DeBERTa reaches F1 0.6476 on the same 5,000-record held-out subset---an absolute gain of +0.475 F1 and a 3.76x relative improvement---and F1 0.6455 on the full 100,002-record test split.** Even the weakest of our three trained models (SC+H with curriculum, F1 0.2772) comfortably exceeds every published comparator. Figure 1 illustrates the gap across all eleven evaluated systems.

The original PIIBench paper characterises the problem. The present work proposes and evaluates a solution. We ask whether a DeBERTa model fine-tuned on the corrected multi-source preparation can

substantially exceed the published comparators, and whether architectural extensions—source conditioning, hierarchical coarse-to-fine prediction, and sequential curriculum learning—improve on simpler direct fine-tuning.

The corrected dataset preparation addresses processing issues identified after the original release: correct Nemotron span parsing, a revised retained label space of 82 entity types, rebalanced Nemotron representation, and corrected train/validation/test membership. Because this updated preparation changes split membership, we re-evaluate all eight comparators on a new reproducible 5,000-record subset drawn exclusively from the corrected test split.

Contributions. This paper makes the following contributions:

- A corrected multi-source PII-Bench preparation with updated Nemotron parsing, an 82-type label space, balanced source mixture, and SHA-256-identified evaluation artifacts.
- Direct DeBERTa fine-tuning as a strong task-specific baseline, achieving span-level F1 0.6455 on the 100,002-record full held-out split—a gain of 0.507 F1 over the strongest published comparator on the same corrected benchmark.
- A source-conditioned hierarchical architecture (SC+H) as a controlled ablation, with entity-level analysis identifying the categories where additional structure helps.
- Empirical evidence that sequential source-family curriculum learning degrades broad-coverage detection via catastrophic forgetting of prior source families.
- A complete reproducible evaluation pipeline with SHA-256-identified test artifacts and code paths made publicly available.

2. Related Work

2.1 Named Entity Recognition Benchmarks

CoNLL-2003 (Sang and De Meulder, 2003) is the canonical NER benchmark, covering four entity types over English and German newswire. OntoNotes 5.0 (Weischedel et al., 2013) extends coverage to 18 types across multiple genres. WikiANN (Pan et al., 2017) provides cross-lingual annotations for 282 languages. MultiNERD (Tedeschi and Navigli, 2022) introduces 15 fine-grained types, and FewNERD (Ding et al., 2021) provides 66 types for episodic learning. None of these benchmarks is designed for PII detection: their taxonomies include non-sensitive types and omit critical PII categories such as financial identifiers and credentials.

2.2 PII-Specific Datasets

The ai4privacy dataset series (ai4privacy, 2023) provides large-scale synthetic PII annotations with 54–63 label classes across multiple languages. The Gretel synthetic finance dataset (Gretel.ai, 2023) covers 29 PII types across 100 financial document formats. NVIDIA’s Nemotron-PII dataset (NVIDIA, 2023) covers 55+ entity types across 50 industries. Each defines an independent label vocabulary, preventing direct cross-dataset evaluation without normalisation.

2.3 Financial NER

FiNER-139 (Loukas et al., 2022) annotates 1.1 million sentences from SEC 10-K/10-Q filings with 139 XBRL financial disclosure tags. While not a classical PII dataset, it provides dense supervision for financial entities underrepresented in general NER corpora.

2.4 PII Detection Systems

Microsoft Presidio (Microsoft, 2023) combines regular-expression patterns with spaCy-backed NLP to identify 18 PII entity types. Transformer-based general NER models—BERT-base NER, XLM-RoBERTa NER, SpanMarker mBERT, SpanMarker BERT—are widely used but trained exclusively on general NER labels. PII-specific models include Piirinha DeBERTa, fine-tuned on ai4privacy-

400k, and XtremeDistil FiNER, fine-tuned on FiNER-139 XBRL tags. All are siloed to their training domain.

2.5 Multi-Domain Adaptation and Curriculum Learning

Sequential curricula can improve data-efficiency when source families have compatible structures, but are prone to catastrophic forgetting (McCloskey and Cohen, 1989) when later phases specialise away from earlier distributions. Our experiments provide direct evidence of this failure mode in the PII setting.

2.6 PII-Bench

Jha (2026) presented the original PII-Bench corpus: 2,369,883 annotated sequences across 48 canonical entity types from ten source datasets, with a normalisation pipeline mapping 80+ source-specific label variants to a standardised BIO scheme. Eight published systems evaluated on the 1,398-record stratified test subset all achieved F1 below 0.14 (best: Presidio at 0.1385). The present work extends PII-Bench with a corrected data preparation and trained model evaluation.

3. Corrected Data Preparation

3.1 Motivation for Correction

Post-release exploratory analysis of the original PII-Bench preparation identified two processing issues: (1) NVIDIA Nemotron PII spans were not parsed correctly from their XML-tagged text format, producing incorrect BIO annotations; (2) Nemotron records were not proportionally balanced, resulting in under-representation. These issues affected train/validation/test membership and motivated a corrected preparation.

3.2 Processing Pipeline

The corrected pipeline applies the following steps in order:

1. Consolidate all retained source datasets to canonical BIO annotations.
2. Re-parse NVIDIA Nemotron PII spans from XML-tagged text and retain only PII-bearing records.
3. Balance Nemotron to approximately 10% of the prepared corpus through stratified sampling.
4. Cap `finer_139` at 150,000 records to prevent financial filings from dominating the mixture.
5. Remove rare entity types with fewer than a configured threshold of B-tag mentions.
6. Produce source-stratified train, validation, and test JSONL splits.

3.3 Prepared Splits

Table 1 reports the resulting split sizes. Training data totals 799,948 records. The 100,002-record test split is the primary evaluation artifact. The `test_5k` subset of 5,000 records is used for multi-system comparative evaluation.

Split	Records	Use
Train	799,948	Model optimisation
Validation	99,990	Held-out validation population
<code>val_1p</code>	994	Fast intra-training model selection
Test	100,002	Final full held-out evaluation
<code>test_5k</code>	5,000	Multi-system comparative evaluation

Table 1: Corrected PII-Bench splits used in this work.

3.4 Label Space and Class Imbalance

After rare-label filtering, 82 entity types are retained. With BIO prefixes and the outside label O, the fine label space contains 165 classes. The hierarchical coarse taxonomy groups these into ten categories plus O, yielding 21 coarse BIO labels. Table 2 summarises the token-level class distribution.

Item	Value
Retained entity types	82
Fine BIO labels (incl. O)	165
Coarse BIO labels (incl. O)	21
Training supervised tokens	52,846,281
O tokens	44,016,269 (83.3%)
Non-O tokens	8,830,012 (16.7%)

Table 2: Label space and token imbalance in the training split.

The 83% outside-token majority causes training collapse under uniform cross-entropy: all models trained without reweighting produced near-zero precision, recall, and F1. All reported trained models therefore use a weighted loss with $w(O) = 0.1$ and $w(\text{entity}) = 1.0$.

3.5 test_5k Evaluation Subset

The `test_5k` artifact is selected from the corrected test split by source-stratified proportional sampling with largest-remainder allocation and random seed 42. The subset contains 222 BIO continuation labels that seqeval interprets as new entity spans. Since every model is evaluated on the identical file with the identical metric, this does not affect relative rankings, but we disclose it as an annotation caveat.

Item	Value
Records	5,000
Gold spans (seqeval)	15,548
Represented entity types	82
Represented sources	10
SHA-256	46238c6cc12526d4d29e9282e96d2936...

Table 3: `test_5k` evaluation subset properties.

4. Methods

We train three DeBERTa-based models of increasing architectural complexity, using `microsoft/deberta-v3-base` (He et al., 2023) as the shared backbone.

4.1 Model A: Direct Fine-Tuned DeBERTa

The simplest model fine-tunes `deberta-v3-base` as a standard BIO token classifier with weighted cross-entropy. No source metadata, auxiliary tasks, or curriculum phases are used. A linear classification head maps each token’s contextualised representation to a distribution over 165 fine BIO labels. This model is released on HuggingFace Hub as `piibench-deberta-base`.

Parameter	Value
Max sequence length	256
Batch size per device	6
Gradient accumulation	10
Effective batch size	60
Learning rate	2e-5
O / entity loss weights	0.1 / 1.0
Validation subset	val_1p, span-level F1
Best validation step	24,000

Table 4: Direct fine-tuning hyperparameters.

4.2 Model B: Source-Conditioned Hierarchical DeBERTa (SC+H)

Released on HuggingFace Hub as `piibench-deberta-sch`.

The first proposed extension adds two components.

A learned dataset-source token is prepended to each input sequence (e.g., `[SRC=ai4privacy]`), conditioning the encoder on data provenance. At inference, `[SRC=general]` is used for out-of-distribution inputs.

A coarse-to-fine hierarchical classifier is applied to the DeBERTa hidden states. The coarse head predicts over ten entity groups; its output probabilities are concatenated with the encoder hidden states before the fine BIO head.

Training optimises a combined loss: $L = L_{\text{fine}} + 0.3 \times L_{\text{coarse}}$.

The ten coarse groups are: PERSON_GROUP, CONTACT, FINANCIAL_ID, TEMPORAL, CREDENTIAL, NETWORK, ORG_ROLE, LOCATION, MISC, and FINANCIAL_NER.

Parameter	Value
Max sequence length	256
Effective batch size	60
Learning rate	2e-5
Coarse loss weight	0.3
O / entity loss weights	0.1 / 1.0
Gradient checkpointing	Enabled
Best validation step	37,000

Table 5: SC+H hyperparameters.

4.3 Model C: SC+H with Curriculum (SC+H+Curr)

The most complex model training uses the Source Conditioned Hierarchical architecture and applies three sequential one-epoch training phases ordered by source family.

Table 6 reports the val_1p F1 at the end of each phase.

Phase	Source Family	val_1p F1
1	General NER	0.131
2	Synthetic PII	0.430
3	Financial PII (<code>gretel_finance</code> , <code>finer_139</code>)	0.305

Table 6: Curriculum training phases and val_1p F1 at phase completion.

F1 improves substantially through Phase 2 but drops following Phase 3, providing direct evidence of catastrophic forgetting: the financial-domain specialisation overwrites the broader PII representations acquired in Phase 2.

5. Experimental Setup

5.1 Training Infrastructure

Field	Value
GPU	1 × NVIDIA L4 (24 GB)
Machine type	g2-standard-12
vCPU / memory	12 vCPUs / 48 GB
Zone	asia-southeast1-c
Mixed precision	BF16
Max sequence length	256 tokens

Table 7: GCP training infrastructure.

Direct DeBERTa training ran for approximately 11.8 hours before best-model export. SC+H completed 39,999 optimiser steps in approximately 17 hours 54 minutes. Full-test inference was run locally on an NVIDIA GeForce RTX 4070 (8 GB VRAM) using a memory-efficient streaming evaluator.

5.2 Evaluation Protocol

All precision, recall, and F1 values use span-level seqeval scoring, requiring exact match of both entity span boundaries and entity-type label. No partial credit is awarded.

The `test_5k` evaluation places all eleven systems on the identical test file. The full 100,002-record evaluation is reserved for the head-to-head competition between Direct DeBERTa and SC+H. To avoid prediction-accumulation memory issues, full-test inference uses a custom streaming evaluator: records are read in chunks of 5,000, inference is batched in smaller minibatches, and seqeval metrics are accumulated incrementally without retaining record-level predictions in memory.

6. Results

6.1 Historical Context: Original PII-Bench Comparators

Table 8 reports the eight comparator systems on the original PII-Bench 1,398-record test subset (Jha, 2026). These scores motivated the present experiments; the corrected `test_5k` scores in Table 9 are the appropriate comparators for the newly trained models.

System	Type	F1	P	R
Microsoft Presidio	Rule-based	0.1385	0.1522	0.1271
spaCy en_core_web_lg	General NER	0.0873	0.0636	0.1395
SpanMarker BERT	General NER	0.0877	0.1723	0.0588
SpanMarker mBERT	General NER	0.0823	0.1477	0.0570
BERT-base NER	General NER	0.0657	0.0791	0.0562
XLM-RoBERTa NER	General NER	0.0494	0.0508	0.0482
Piirinha DeBERTa	PII-specific	0.0463	0.0604	0.0375
XtremeDistil FiNER	Financial NER	0.0413	0.6990	0.0213

Table 8: Original PII-Bench comparators on the 1,398-record test subset (Jha, 2026).

6.2 Corrected test_5k Complete Comparison

Table 9 reports all eleven systems on the corrected test_5k subset. Direct DeBERTa achieves F1 0.6476, an absolute gain of 0.475 over the strongest published comparator (SpanMarker BERT at 0.1723). The curriculum variant at 0.2772 still outperforms every published comparator by at least 0.105 F1, demonstrating that even imperfect multi-source fine-tuning substantially widens the gap over domain-siloed systems.

System	Type	F1	P	R
Direct DeBERTa	Fine-tuned	0.6476	0.6300	0.6662
SC+H	Proposed	0.5899	0.5565	0.6274
SC+H+Curr	Proposed+Curr	0.2772	0.3491	0.2299
SpanMarker BERT	General NER	0.1723	0.4266	0.1080
SpanMarker mBERT	General NER	0.1622	0.3659	0.1042
Microsoft Presidio	Rule-based	0.1421	0.1748	0.1197
BERT-base NER	General NER	0.1321	0.1834	0.1032
XLM-RoBERTa NER	General NER	0.1153	0.1373	0.0993
Piirinha DeBERTa	PII-specific	0.0872	0.1204	0.0684
spaCy en_core_web_lg	General NER	0.0853	0.0677	0.1152
XtremeDistil FiNER	Financial NER	0.0191	0.6089	0.0097

Table 9: All systems on the corrected test_5k subset. Bold = best per metric.

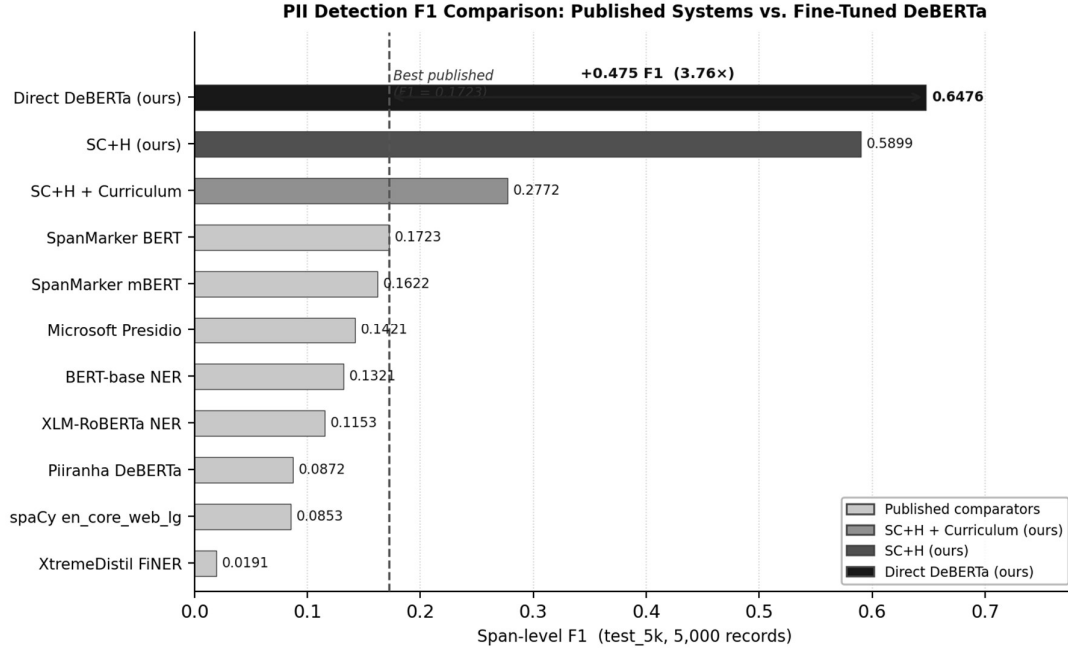


Figure 1: Span-level F1 on test_5k (5,000 records) for all eleven evaluated systems. Light grey = published comparators; dark bars = fine-tuned DeBERTa models. Dashed line marks the best published system (SpanMarker BERT, F1 = 0.1723). Direct DeBERTa achieves F1 0.6476 — a **3.76x improvement and +0.475 absolute F1 gain** over the best published comparator.

6.3 Validation–Test Disagreement and Final Full-Test Evaluation

The fast val_1p subset showed a marginal advantage for SC+H (F1 0.661 vs. 0.645 for Direct DeBERTa). The test_5k result reversed this, favouring Direct DeBERTa by 0.058. To resolve the disagreement, both models were evaluated on the complete 100,002-record test split.

Evaluation	Direct F1	SC+H F1	Gap	Winner
val_1p	0.6445	0.6612	+0.017 SC+H	SC+H
test_5k	0.6476	0.5899	+0.058 Direct	Direct

Table 10: Validation–test disagreement between Direct DeBERTa and SC+H.

Model	Records	F1	P	R
Direct DeBERTa	100,002	0.6455	0.6277	0.6645
SC+H	100,002	0.5894	0.5560	0.6270
<i>Direct – SC+H</i>	—	+0.0561	+0.0717	+0.0375

Table 11: Final full-test evaluation (100,002 records, 308,421 gold spans). Bold = best per metric.

Direct DeBERTa is superior on all three metrics across the full held-out distribution. This is the primary result: the simplest fine-tuning approach provides the strongest broad-coverage PII detection.

7. Entity-Level Analysis

7.1 Overall Type-Level Winner Counts

Of the 82 retained fine entity types, Direct DeBERTa achieves higher F1 on 54 types and SC+H on 28 types, with no ties. Direct DeBERTa wins all ten coarse groups by support-weighted mean fine-entity F1.

7.2 Coarse Group Summary

Table 12 reports support-weighted mean F1 per coarse group on the full test split. Direct DeBERTa wins every group. The largest advantage is on FINANCIAL_ID, where direct fine-tuning wins all 11 constituent entity types and holds a 0.146 F1 advantage—despite the curriculum variant having ended with financial-domain specialisation.

Group	Support	Direct F1	SC+H F1	Δ	Wins D/S
FINANCIAL_NER	58,821	0.3229	0.2412	+0.082	1/0
LOCATION	53,111	0.7151	0.6729	+0.042	6/4
PERSON_GROUP	46,789	0.8004	0.7515	+0.049	5/1
ORG_ROLE	30,723	0.7422	0.7252	+0.017	2/3
TEMPORAL	30,683	0.5923	0.5548	+0.038	2/2
NETWORK	24,406	0.6611	0.5920	+0.069	5/4
MISC	23,574	0.7318	0.6321	+0.100	11/5
CONTACT	18,437	0.7087	0.6593	+0.049	2/2
CREDENTIAL	12,882	0.8902	0.8611	+0.029	9/7
FINANCIAL_ID	8,995	0.8763	0.7305	+0.146	11/0

Table 12: Coarse group results on the full test split. $\Delta = \text{Direct F1} - \text{SC+H F1}$.

7.3 Entity Types Most Favouring Direct Fine-Tuning

Entity	Group	Supp.	Direct	SC+H	Δ
CRYPTO_ADDRESS	MISC	1,569	0.8641	0.0012	+0.863
VEHICLE	MISC	591	0.9787	0.4689	+0.510
IBAN	FINANCIAL_ID	769	0.8695	0.4952	+0.374
ACCOUNT_NUMBER	FINANCIAL_ID	2,686	0.9095	0.6505	+0.259
PHONE	CONTACT	1,535	0.9954	0.8213	+0.174
IP_ADDRESS	NETWORK	6,178	0.5528	0.4198	+0.133
SSN	CREDENTIAL	1,261	0.9517	0.8261	+0.126
NAME	PERSON_GROUP	6,645	0.5464	0.4355	+0.111
USERNAME	NETWORK	8,287	0.5285	0.4185	+0.110
FINANCIAL_ENTITY	FINANCIAL_NER	58,821	0.3229	0.2412	+0.082

Table 13: Top 10 entity types by Direct DeBERTa advantage on the full test split.

7.4 Entity Types Most Favouring SC+H

Entity	Group	Supp.	Direct	SC+H	Δ
HTTP_COOKIE	NETWORK	595	0.1812	0.5752	+0.394
LOCAL_LATLNG	LOCATION	55	0.3810	0.4655	+0.085
BLOOD_TYPE	MISC	462	0.8265	0.8862	+0.060
DATE_TIME	TEMPORAL	1,463	0.7422	0.7844	+0.042
COUNTY	LOCATION	611	0.8980	0.9275	+0.030
COORDINATE	LOCATION	857	0.8874	0.9068	+0.020
EDUCATION_LEVEL	MISC	1,137	0.7623	0.7799	+0.018
PHONE_NUMBER	CONTACT	3,079	0.8606	0.8738	+0.013
STATE	LOCATION	1,974	0.8291	0.8384	+0.009
COMPANY_NAME	ORG_ROLE	8,674	0.8487	0.8560	+0.007

Table 14: Top 10 entity types by SC+H advantage on the full test split.

SC+H's largest advantage is for HTTP_COOKIE (+0.394), a contextually complex type where source provenance information may provide a useful disambiguation signal. These SC+H gains are concentrated on lower-support types and are outweighed by the direct model's advantages on high-frequency categories such as FINANCIAL_ENTITY, IP_ADDRESS, USERNAME, ACCOUNT_NUMBER, PHONE, and SSN.

8. Discussion

8.1 Simplicity Wins on Broad Coverage

The primary result is that a directly fine-tuned DeBERTa model on broad, correctly prepared multi-source PII data substantially outperforms all tested comparators and both architectural extensions. The F1 improvement over the strongest published system (0.475 on test_5k) is an order of magnitude larger than the difference between Direct DeBERTa and SC+H (0.058 on test_5k, 0.056 on full test). The data preparation quality and the weighted loss function account for more of the performance gain than the architectural choices.

8.2 Architectural Complexity and Localised Trade-offs

SC+H is not uniformly weaker. Its source-conditioned hierarchical representations produce meaningful F1 gains on 28 of 82 entity types, particularly contextually grounded types such as HTTP_COOKIE and DATE_TIME. A hybrid system that selectively applies hierarchical decoding for certain coarse categories while using direct decoding for others could potentially recover these localised gains without incurring the broad-coverage regression.

8.3 Catastrophic Forgetting in Sequential Curriculum

The curriculum variant provides a clear illustration of catastrophic forgetting: after Phase 2's substantial gain (0.430 on val_1p), Phase 3 financial specialisation drops performance to 0.305. Despite ending with financial-domain training, SC+H+Curr is substantially weaker than Direct DeBERTa on FINANCIAL_ID types. The curriculum schedule tested (General NER $\rightarrow$ Synthetic PII $\rightarrow$ Financial PII) is one of many possible orderings; the result demonstrates that naive sequential specialisation is harmful but does not exclude all curriculum designs.

8.4 Validation–Test Reversal

The val_1p model selection metric marginally preferred SC+H (0.661 vs. 0.645), but the larger test_5k and full test evaluations reversed this by 0.058 and 0.056 respectively. The reversal indicates that the 994-record fast-validation subset carries insufficient statistical resolution to distinguish models in the 0.01–0.02 F1 range.

8.5 Operational Implications

For practitioners deploying PII detection on heterogeneous text, the results support a straightforward approach: collect broad, multi-domain annotated data, correct any source-specific parsing issues, apply weighted cross-entropy to handle class imbalance, and fine-tune a strong pretrained encoder directly. Architectural elaboration should be motivated by demonstrated failure modes on specific entity categories rather than assumed a priori.

9. Limitations and Future Work

Dataset preparation change. The corrected preparation differs from the original PII-Bench paper’s preparation. Trained models must therefore be compared with the rerun comparators on test_5k rather than directly with the original Table 5 numbers.

BIO continuation caveat. The test_5k artifact contains 222 BIO continuation labels that seqeval treats as new spans. All systems are affected equally for relative ranking. A BIO-normalised sensitivity analysis is deferred to future work.

Single curriculum schedule. The curriculum experiment tests one specific ordered source-family schedule. The negative result does not preclude all curriculum designs.

Single training seed. No repeated-seed training runs were performed. Architecture effects cannot be separated from checkpoint variance without multi-seed replication.

No statistical confidence intervals. Bootstrap confidence intervals on full-test predictions are not computed. Formal significance testing is deferred.

English only; domain coverage gaps. PII-Bench covers general, synthetic PII, and financial domains but excludes medical records, legal contracts, and social media. Extending coverage and adding multilingual evaluation tracks are planned directions.

10. Conclusion

We have presented a fine-tuning study on PII-Bench, a corrected multi-source PII detection benchmark spanning 82 entity types and ten heterogeneous source datasets. Against eight published comparator systems—rule-based, general NER, PII-specific, and financial NER—a directly fine-tuned DeBERTa model achieves F1 0.6455 on the 100,002-record full held-out split, an absolute gain of more than 0.50 F1 over the strongest published comparator.

A source-conditioned hierarchical extension (SC+H) is competitive but consistently weaker across the full held-out distribution, winning only 28 of 82 entity types. A sequential curriculum variant (SC+H+Curr) degrades substantially due to catastrophic forgetting in its financial-domain final phase.

The principal finding is that broad-coverage PII detection benefits more from correct multi-source data preparation and a simple weighted training objective than from the tested architectural and curriculum complexity. SC+H retains localised advantages on 28 entity types where source conditioning and coarse supervision appear beneficial, warranting further investigation.

The code, evaluation artifacts, and trained model checkpoints are available at <https://github.com/pritesh-2711/pii-bench>. The dataset is available at <https://huggingface.co/datasets/pritesh-2711/pii-bench>. Fine-tuned models are released on HuggingFace Hub: `piibench-deberta-base` (direct fine-tune) and `piibench-deberta-sch` (SC+H).

References

- ai4privacy. (2023). PII Masking 400K / 300K. HuggingFace Hub. Custom academic licence.
- Bengio, Y., Louradour, J., Collobert, R., and Weston, J. (2009). Curriculum learning. *Proceedings of ICML*, 41–48.
- Ding, N., et al. (2021). Few-NERD: A few-shot named entity recognition dataset. *ACL-IJCNLP 2021*.
- Gretel.ai. (2023). Synthetic PII Finance Multilingual Dataset. HuggingFace Hub: [gretelai/synthetic_pii_finance_multilingual](https://huggingface.co/gretelai/synthetic_pii_finance_multilingual).
- He, P., Gao, J., and Chen, W. (2023). DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. *ICLR 2023*.
- Honnibal, M. and Montani, I. (2017). spaCy 2: Natural language understanding with Bloom embeddings, convolutional neural networks and incremental parsing.
- Isotonic. (2023). PII Masking 200K. HuggingFace Hub: [Isotonic/pii-masking-200k](https://huggingface.co/Isotonic/pii-masking-200k). Apache 2.0 Licence.
- Jha, P. (2026). PII-Bench: A unified multi-source benchmark corpus for personally identifiable information detection. *arXiv:2604.15776*.
- Loukas, L., et al. (2022). FiNER: Financial numeric entity recognition for XBRL tagging. *ACL 2022*.
- McCloskey, M. and Cohen, N. J. (1989). Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24:109–165.
- Microsoft. (2023). Presidio — Data Protection SDK. <https://github.com/microsoft/presidio>.
- NVIDIA. (2023). Nemotron-PII: A synthetic dataset for PII detection. HuggingFace Hub: [nvidia/Nemotron-PII](https://huggingface.co/nvidia/Nemotron-PII). CC-BY 4.0 Licence.
- Pan, X., et al. (2017). Cross-lingual name tagging and linking for 282 languages. *ACL 2017*.
- Sang, E. F. and De Meulder, F. (2003). Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. *CoNLL 2003*.
- Tedeschi, S. and Navigli, R. (2022). MultiNERD: A multilingual, multi-genre and fine-grained dataset for named entity recognition. *NAACL Findings 2022*.
- Weischedel, R. et al. (2013). OntoNotes Release 5.0. LDC Catalog No.: LDC2013T19.
- Wolf, T., et al. (2020). Transformers: State-of-the-art natural language processing. *EMNLP 2020 System Demonstrations*.

Appendix A: Complete Fine-Entity Full-Test Comparison

All 82 retained fine entity types on the complete 100,002-record held-out split. Δ = Direct F1 – SC+H F1. D = Direct DeBERTa wins; S = SC+H wins.

Entity	Group	Supp.	Dir.F1	SC+H	Δ	W
ACCOUNT_NUMBER	FINANCIAL_ID	2,686	0.9095	0.6505	+0.259	D
BBAN	FINANCIAL_ID	102	0.5769	0.5202	+0.057	D
BIC	FINANCIAL_ID	231	0.9360	0.8148	+0.121	D
CREDIT_CARD	FINANCIAL_ID	2,305	0.9115	0.8088	+0.103	D
CREDIT_CARD_NUMBER	FINANCIAL_ID	95	0.5422	0.5287	+0.013	D
CREDIT_DEBIT_CARD	FINANCIAL_ID	899	0.8615	0.8568	+0.005	D
CVV	FINANCIAL_ID	362	0.8537	0.8360	+0.018	D
IBAN	FINANCIAL_ID	769	0.8695	0.4952	+0.374	D
SWIFT_BIC	FINANCIAL_ID	592	0.8423	0.8189	+0.023	D
SWIFT_BIC_CODE	FINANCIAL_ID	122	0.6324	0.5172	+0.115	D
BANK_ROUTING_NUMBER	FINANCIAL_ID	832	0.8214	0.8005	+0.021	D
ADDRESS	LOCATION	6,490	0.6179	0.5269	+0.091	D
CITY	LOCATION	2,036	0.8512	0.8324	+0.019	D
COORDINATE	LOCATION	857	0.8874	0.9068	−0.020	S
COUNTRY	LOCATION	2,887	0.8931	0.8642	+0.029	D
COUNTY	LOCATION	611	0.8980	0.9275	−0.030	S
LOC	LOCATION	32,846	0.6879	0.6441	+0.044	D
LOCAL_LATLNG	LOCATION	55	0.3810	0.4655	−0.085	S
POSTCODE	LOCATION	798	0.9021	0.8652	+0.037	D
STATE	LOCATION	1,974	0.8291	0.8384	−0.009	S
STREET_ADDRESS	LOCATION	4,557	0.7411	0.7154	+0.026	D
AGE	PERSON_GROUP	454	0.8427	0.8447	−0.002	S
FIRST_NAME	PERSON_GROUP	6,315	0.9326	0.9003	+0.032	D
GENDER	PERSON_GROUP	509	0.9118	0.9082	+0.004	D
LAST_NAME	PERSON_GROUP	4,684	0.9289	0.8777	+0.051	D
NAME	PERSON_GROUP	6,645	0.5464	0.4355	+0.111	D
PERSON	PERSON_GROUP	28,182	0.8066	0.7674	+0.039	D
COMPANY	ORG_ROLE	4,039	0.5686	0.5688	−0.000	S
COMPANY_NAME	ORG_ROLE	8,674	0.8487	0.8560	−0.007	S

JOB	ORG_ROLE	2,645	0.9249	0.9129	+0.012	D
OCCUPATION	ORG_ROLE	3,136	0.4726	0.4808	−0.008	S
ORG	ORG_ROLE	12,229	0.7537	0.7061	+0.048	D
DATE	TEMPORAL	18,080	0.6721	0.6215	+0.051	D
DATE_OF_BIRTH	TEMPORAL	1,465	0.9497	0.9540	−0.004	S
DATE_TIME	TEMPORAL	1,463	0.7422	0.7844	−0.042	S
TIME	TEMPORAL	9,675	0.3665	0.3348	+0.032	D
DEVICE_IDENTIFIER	NETWORK	496	0.8872	0.8774	+0.010	D
HTTP_COOKIE	NETWORK	595	0.1812	0.5752	−0.394	S
IP_ADDRESS	NETWORK	6,178	0.5528	0.4198	+0.133	D
IPv4	NETWORK	1,042	0.8395	0.8448	−0.005	S
IPv6	NETWORK	591	0.8421	0.8459	−0.004	S
MAC_ADDRESS	NETWORK	606	0.8397	0.8412	−0.002	S
URL	NETWORK	5,319	0.9003	0.8663	+0.034	D
USERNAME	NETWORK	8,287	0.5285	0.4185	+0.110	D
USER_NAME	NETWORK	1,292	0.8685	0.8598	+0.009	D
AMOUNT	MISC	872	0.4033	0.2444	+0.159	D
BLOOD_TYPE	MISC	462	0.8265	0.8862	−0.060	S
CC_SECURITY_CODE	MISC	81	0.5920	0.2897	+0.302	D
CRYPTO_ADDRESS	MISC	1,569	0.8641	0.0012	+0.863	D
CURRENCY	MISC	1,931	0.8878	0.8290	+0.059	D
EDUCATION_LEVEL	MISC	1,137	0.7623	0.7799	−0.018	S
EMPLOYMENT_STATUS	MISC	720	0.8789	0.8571	+0.022	D
LANGUAGE	MISC	669	0.8457	0.8209	+0.025	D
LICENSE_PLATE	MISC	719	0.9444	0.9474	−0.003	S
MISC	MISC	12,092	0.6334	0.5973	+0.036	D
POLITICAL_VIEW	MISC	383	0.8370	0.8287	+0.008	D
RACE_ETHNICITY	MISC	750	0.8541	0.8538	+0.000	D
RELIGIOUS_BELIEF	MISC	452	0.8443	0.8502	−0.006	S
SEXUALITY	MISC	230	0.9065	0.8993	+0.007	D
VEHICLE	MISC	591	0.9787	0.4689	+0.510	D
VEHICLE_IDENTIFIER	MISC	916	0.9467	0.9494	−0.003	S

EMAIL	CONTACT	13,117	0.6373	0.5848	+0.053	D
FAX_NUMBER	CONTACT	706	0.7504	0.7568	−0.006	S
PHONE	CONTACT	1,535	0.9954	0.8213	+0.174	D
PHONE_NUMBER	CONTACT	3,079	0.8606	0.8738	−0.013	S
ACCOUNT_PIN	CREDENTIAL	91	0.5890	0.3333	+0.256	D
API_KEY	CREDENTIAL	693	0.8167	0.8103	+0.007	D
BIOMETRIC_ID	CREDENTIAL	1,090	0.9078	0.9082	−0.000	S
CERT_LICENSE_NUM	CREDENTIAL	760	0.9122	0.9056	+0.007	D
CUSTOMER_ID	CREDENTIAL	1,882	0.8876	0.8887	−0.001	S
DRIVER_LICENSE	CREDENTIAL	80	0.6550	0.6447	+0.010	D
EMPLOYEE_ID	CREDENTIAL	1,334	0.8669	0.8499	+0.017	D
HP_BENEF_NUMBER	CREDENTIAL	640	0.8915	0.8941	−0.003	S
MEDICAL_RECORD	CREDENTIAL	734	0.9658	0.9666	−0.001	S
NATIONAL_ID	CREDENTIAL	300	0.8534	0.8561	−0.003	S
PASSPORT_NUMBER	CREDENTIAL	115	0.6772	0.6190	+0.058	D
PASSWORD	CREDENTIAL	2,510	0.9070	0.8655	+0.042	D
PIN	CREDENTIAL	937	0.8686	0.8054	+0.063	D
SSN	CREDENTIAL	1,261	0.9517	0.8261	+0.126	D
TAX_ID	CREDENTIAL	252	0.8643	0.8683	−0.004	S
UNIQUE_ID	CREDENTIAL	203	0.8064	0.8276	−0.021	S
FINANCIAL_ENTITY	FINANCIAL_NER	58,821	0.3229	0.2412	+0.082	D

Table A1: Full entity-level comparison on the 100,002-record test split. W: D=Direct wins, S=SC+H wins.